

Mauro Bernardini

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Research Activity

Postdoctoral Researcher in Data Science and Astrophysics

University of Zurich
December 2023 - present

Scientific referent: Robert Feldmann

Research Focus: Structure and galaxy formation and evolution, Machine and Deep Learning

- Focus on Deep Learning, Galaxy formation and evolution, Active Galactic Nuclei physics, accretion and feedback
- My current projects involve running galaxy formation simulations and train neural networks on their output

Main contributions: During my first postdoc I have worked on projects that deploy Deep Learning to answer astrophysical questions, e.g. how dark matter and cosmic gas (e.g. HI) are connected. I have developed a new algorithm termed EMBER-2 that uses Neural Networks to paint baryons onto dark matter only simulations at the field level (Bernardini et al. 2025a). I have further used EMBER-2 to build a model that can predict the underlying dark matter distribution from HI observations, which could be important for future data analysis pipelines (Bernardini et al. 2025b).

Ph.D. in Data Science and Astrophysics

University of Zurich
October 2019 - 2023

Scientific referent: Robert Feldmann, Lucio Mayer, Jan Dirk Wegner

Thesis title: Emulating Cosmic Gas with generative Deep Learning

- Focus on Deep Learning, interface between simulations and observations, Dark Matter, cosmic gas and HI

Main contributions: In my first PhD project I have developed a neural network based algorithm to identify lagrangian patches in the initial conditions of numerical simulations, that will form halos at present time (Bernardini et al. 2020). During later work in my PhD I focused on numerical simulations of galaxy formation with a special focus on cosmic gas, and especially HI. I developed the first iteration of EMBER, a Neural Network based field level emulator, that predicts cosmic gas distributions from dark matter inputs (Bernardini et al. 2022).

Teaching and Mentorship

Teaching:

- Introduction to Data Science — 2020, 2023, 2024
- Introduction to Astrophysics — 2022
- Computational Astrophysics — 2021
- Astronomy Field Trips — 2019, 2020, 2021, 2022

Mentorship:

- Co-supervision: Lucas Tortora, 2021 - 2023 (Bachelor thesis)
  The HI covering fraction of Lyman Limit Systems in FIRE haloes (Tortora et al. 2024)

- Co-supervision: Andrin Rehmann, 2024 - 2025 (Master thesis)
  Predicting Dark Matter Initial Conditions

Publications

- Published papers: 11 (4 first author, 7 co-author)
- Submitted papers: 2 (1 first and 1 co-author)
- In preparation: 3
- h-index 6

Grants and Achievements

Production project on LUMI-C
Zurich
*PI: **Mauro Bernardini**, Co-PI: Robert Feldmann*
April 2025 - 2026

Project Name: Constraining the physics of Active Galactic Nuclei in massive galaxies with cosmological FIRE simulations and Machine Learning
- This computational production project is the direct continuation of the previous CSCS project
- We were granted a total of **180,000 node hours** over 1 year on the LUMI-C multicore supercomputer

Production project on CSCS
Zurich
*PI: **Mauro Bernardini**, Co-PI: Robert Feldmann*
September 2023 - 2024

Project Name: FIREbox-AGN: Constraining the physics of Active Galactic Nuclei with cosmological simulations and Machine Learning
- This project was designed to train our Machine Learning framework EMBER-2 on a large suite of cosmological volume simulations with varying AGN physics
- We were granted a total of **168,000 node hours** over 1 year on the Eiger@Alps multicore supercomputer

Research highlight in Nature Astronomy
Honorable mention: The EMBER-2 paper got featured as a research highlight in Nature Astronomy (Oldham 2025)

Education

Master degree in Astrophysics
University of Zurich
Thesis supervisor: Lucio Mayer
2017 - 2019

Thesis title: Thinking inside the box: Machine Learning applications in cosmological structure formation
- Focus on Deep Learning applied to cosmological simulations to identify dark matter haloes

Bachelor degree in Physics
University of Zurich
Thesis supervisor: Prasenjit Saha
2013 - 2017

Thesis title: Characterization of the local velocity field with TGAS-RAVE
- Focus on computation of galactic dynamics from a subset of the GAIA catalogue to identify new stellar streams